

Module 3 Reflective Journal

ITAI 1371 – Machine Learning

Neural Crew

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This lab helped me understand machine learning better than I did before. I had heard terms like supervised learning, unsupervised learning, reinforcement learning, and Logistic Regression before, but I had not really worked with them much. Actually going through the notebook and running the code made everything feel more real. Instead of only reading about machine learning, I got to see how data can be loaded, analyzed, graphed, and used to make predictions.

At the beginning of the lab, I looked at the first step where we had to import all of the essential libraries. I was honestly a little overwhelmed because there were several lines of code, and I did not know what every library was doing right away. At first, it looked like a lot to take in. I decided to slow down and go through it line by line instead of trying to understand everything at once. That helped my understanding a lot. I started to see that each library has a purpose, such as working with data, making graphs, or creating and testing the machine-learning model.

It was really cool seeing what coding can actually do when all of those libraries work together. I am still learning Python, so sometimes code can look confusing at first, especially when there are several lines in one cell. But when I ran the cells and saw the output underneath, it helped connect the code to what it was doing. I liked that Google Colab allowed me to run the code a little at a time and see the results as I worked. It made the assignment feel less like memorizing information and more like learning how the process actually works.

One part I enjoyed the most was making the graphs and analyzing the dataset. The graphs made the information easier to understand because I could actually see patterns in the Wine dataset instead of only looking at numbers in a table. Looking at the data this way helped me understand why data analysis is important before building a model. If someone does not understand the data they are working with, it would be hard to know what features may be useful or what results make sense.

I also learned more about supervised, unsupervised, and reinforcement learning. Supervised learning made the most sense to me because it uses labeled data, meaning the model has examples with known answers that it can learn from. Unsupervised learning is different because it looks for patterns or groups in data without being told the correct answers ahead of time. Reinforcement learning seems more like learning through trial and error, where an agent makes choices and receives rewards or penalties. Understanding the differences between these

types of learning helped me see that machine learning is not just one thing. Different methods are used depending on the type of problem someone is trying to solve.

Towards the end of the lab, I started getting confused when I was working with the model accuracy. I was not completely sure why the accuracy could change or what I was supposed to expect when testing the model. At first, I thought that if my accuracy was lower than the original model, I must have done something wrong. After I slowed down and looked at what I changed, I realized that using different features can change the outcome of the model. Some features may give the model more useful information than others, so the accuracy can go up or down depending on what is selected. Once I understood that, the results made a lot more sense.

That was probably one of the biggest things I learned from this assignment. A machine-learning model is not just going to give the same result no matter what. The data and features that are chosen matter. Changing the input features can affect how well the model predicts the correct class. Seeing that happen in the notebook helped me understand feature selection better than just reading about it would have.

I also enjoyed learning about Logistic Regression. Before this lab, the name sounded more complicated than it does now. I am still not fully comfortable with all of the math behind it, but seeing the code run and seeing the model make predictions helped me understand the basic idea. The Logistic Regression model used information from the Wine dataset to predict the different wine classes. It was interesting to see that a computer can use patterns in data to make predictions, but that the model's results still depend on the data and features that are used.

Overall, I enjoyed this lab because it gave me more experience with coding and showed me more of what can be done with Python and machine learning. I especially liked making the graphs, analyzing the dataset, and seeing how changing features affected the model accuracy. Some parts were confusing, especially near the end when I was trying to understand the accuracy results, but working through that confusion helped me learn more. I am still new to machine learning and coding, but I feel more interested in Logistic Regression now and excited to learn more about it. I also want to get better at understanding the code, choosing useful features, and learning how to improve a model's accuracy in future assignments.